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Subject Code:	MAP711S
Subject Name:	Mobile Application Development
Test Duration:	90 Minutes
Total Marks:	50 Marks
Student Name:	
Student Number:	

Test (Main)

Section A: Multiple Choice [8 marks]

The correct answers are in bold.

1. Which function defines a layout in your app using Composable functions?

   A. **setContent()**
   B. main()
   C. onCreate()
   D. onEntry()

2. What is always returned by Composable functions?

   A. String
   B. another function
   C. **nothing**
   D. it varies, depends on the developer

3. A Composable that has a background color and contains other Composables is called a _____.

   A. color
   B. Container
   C. **Surface**
   D. Box

4. To enable USB debugging on your device, you will tap the build number __ times.

   A. 3

B. 5

C. **7**

D. 9

5. By default, the unit used for Android font is _____.

   A. dp

   B. **sp**

   C. inch

   D. cm

6. A theme cannot be applied on a(n) _____.

   A. View Hierarchy

   B. Activity

   C. entire app

   D. **View**

7. What does this program do?

```
interface A {
   fun print() { println("A") }
}
interface B {
   fun print() { println("B") }
}
abstract class C {
   open fun print() { println("C") }
}
abstract class D {
   open fun print() { println("D") }
}

class Example: C(), D(), A, B {
   override fun print() {
       println("Example")
   }
}

fun main() {
   Example().print()
}
```

   A. prints "Example"

   B. prints "A"

   C. prints "B"

   D. prints "C"

   E. prints "D"

   F. **throws a compilation error**

   G. prints nothing

8. What does the following program do?

```
val oddNumbers = MutableList(5) { i → i * 2 + 1}
val it = oddNumbers.listIterator()
while (it.hasNext()) {
   val value = it.next()
   if (value % 3 == 0) {
       it.remove()
   }
   it.add(value + 1)
}
print(oddNumbers)
```

A. prints 1 2 3 4 5 6 7 8 9 10

B. **prints 1 2 4 5 6 7 8 10**

C. prints 1 3 5 7 9

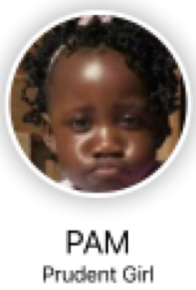
D. prints 2 4 6 8 10

G. nothing because of a compilation error

Section B: Structured Questions [12 marks]

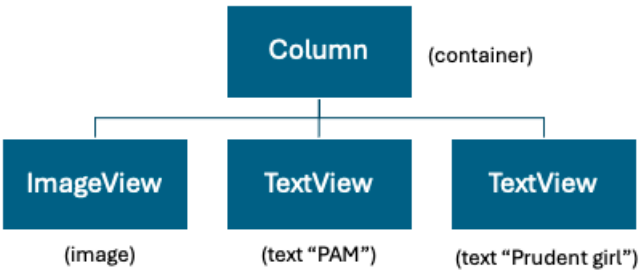
Sample answers are in italics. Kindly accept other meaningful solutions.

1. Briefly explain the Android UI elements in the graphic below, describing its organisation. **[5 marks]**



Three views are contained in a column.
The topmost view is an Image view, with two TextField views below it.
The image has a circular modifier,
and the first TextField has the characters "PAM",
the other text field has the characters "Prident Girl".

Also, draw the element's hierarchy structure and justify its organisation/appearance.**[5 marks]**



2. To handle user text input, Android uses ***textField*** or ***basicTextField*** implementation. **[1 mark]** *//accept any one*

3. RecyclerView is usually equated to Composable's ***LazyRow*** or ***LazyColumn*** **[1 mark]** *//accept any one*

Section C: Open Ended Questions [30 marks]

Sample answers are in italics. Kindly accept other meaningful solutions.

1. Despite Android's success in having the most significant number of users in the world, explain the possible challenges developers face and how they may overcome them. **[5 marks]**
   Fragmentation: *There exists several platforms on which Android applications can run, making it difficult to develop applications that match all platform requirements. This has been made so by the Android Open Source Project which allows hardware manufacturers to port source code and customise it to their liking. Developers can overcome this challenge by specifying target platforms using parameters such as minimum SDK in their settings, making use of support libraries such as AndroidX, and testing their apps on multiple platforms. (2.5 marks)*
   Developer Competition: *Android has the most applications on Play store, meaning there exists a high number of developers competing for the same market share. As a developer, there is need to continually update oneself with fast-changing APIs and good practices to remain current when developing mobile applications. This will ensure favourable user experiences which usually yields good financial returns as the user base increases. (2.5 marks)*

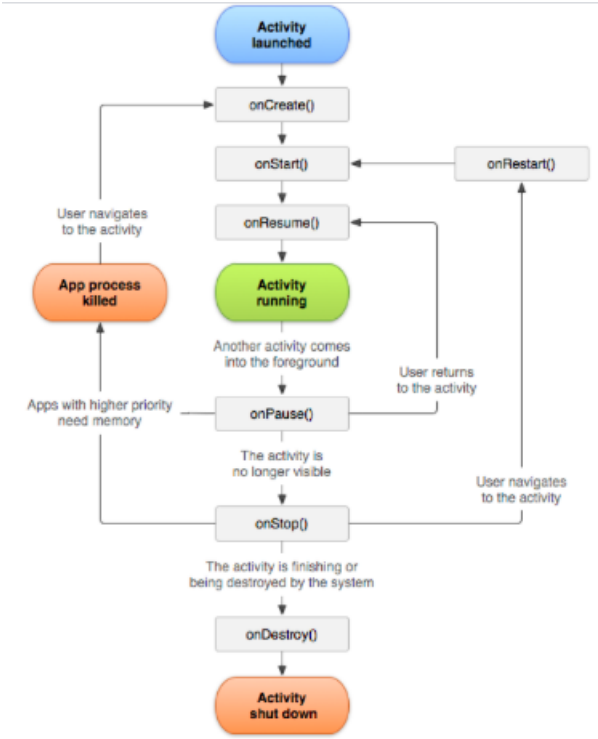
2. With an example, briefly explain state hoisting. **[5 marks]**
   A state represents any data displayed on the screen that can change over time. Instead of keeping that data (state) in a single composable, it is lifted to the parent composable. This concept, known as state hoisting, makes it possible for the single source of truth as the state is shared across child composables,thus promoting reusability in the process.Typically, state hoisting is used to separate UI logic and state management, making it easier to maintain the code.
   Accept any example that separates UI logic from state managment.

3. Using practical examples, explain the benefit of generics in coding. **[10 marks]**
   Generics promotes code reuse, making the code easier to read and maintain. A typical example is when sorting; one does not have to overload functions if there are different data types to be sorted. Below code will be sufficient:

```
fun <T : Comparable<T>> quickSort(collection: Collection<T>): Collection<T> { ... }

quickSort(listOf(1.0, 2.0, 3.0)) // OK

quickSort(listOf(1, 2, 3)) // OK

quickSort(listOf("one", "two", "three")) // OK
```

When using generics, the compiler knows the exact type of data to retrieve without explicitly typecasting. This cakes coding a bit faster and also improves efficiency. In the same vein, generics promotes type safety since type-checking is enforced at compilation time.
Accept any meaningful examples.

4. With the aid of a diagram and real-world use cases, explain the Android Activity Lifecycle. **[10 marks]**



Accept any meaningful corresponding explanation for 5 marks, diagram 5 marks.

This paper consists of 5 pages (including the front page).